Warehouse Safety & Operational Performance Analysis

Independent practice project | SQL and Microsoft Excel

Objective

Analyse illustrative warehouse incident records to assess incident frequency, severity distribution and lost-day impact, then summarise findings for management.

Dataset and cleaning

The source dataset has 12 rows and six fields: Incident ID, Warehouse, Severity, Incident Type, Orders and Lost Days. It contains one duplicate Incident ID (1005) and inconsistent severity casing. After removing the duplicate and standardising severity to uppercase, 11 records remain.

Tools and methods

SQL: filtering, GROUP BY, aggregations, CASE expressions and ROW_NUMBER for duplicate handling.
Excel: data cleaning, formulas, PivotTables, PivotCharts, Power Query practice and dashboard visualisation.

Metric definitions

Metric	Definition
Incident rate	Warehouse incident count / warehouse-level orders × 100
High severity %	High incidents / total incidents × 100
Total lost days	Sum of lost days for the warehouse
Lost days per incident	Total lost days / total incidents

Warehouse KPI results

Warehouse	Incidents	High %	Lost days	Orders	Incident rate	Lost days / incident
BLR1	3	66.67%	15	1000	0.300%	5.00
CHN1	3	66.67%	23	800	0.375%	7.67
HYD1	3	33.33%	11	1200	0.250%	3.67
MAA1	2	0.00%	5	900	0.222%	2.50

Key findings

CHN1 has the highest incident rate (0.375%), total lost days (23), and lost days per incident (7.67). BLR1 and CHN1 both have the highest high-severity share at 66.67%. CHN1 is a priority for further investigation; BLR1 also warrants attention due to its high-severity share.

Recommended investigation

Review incident types, work shifts, equipment conditions, workload and training records. Validate metric definitions and denominator assumptions with stakeholders, then monitor incident rate, severity share and lost days over time.

Limitations and transparency

This is an independent practice project using a small illustrative dataset, not actual Amazon data or a production system. Orders is repeated on incident rows as a warehouse-level denominator; calculations use one warehouse-level value rather than summing repeated values. The data does not establish root causes or statistically robust trends.